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Introduction to Tribology Winter 2026

Project Report on
Analysis of Static Friction Behaviour During Stick-Slip Motion

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Abstract

This study investigates the coefficient of static friction under stick-slip motion using a simple spring–block experimental setup. Different materials including **Titanium (6AL-4V)**, **Stainless Steel**, **Polycarbonate**, **Inconel 625**, and **Aluminium (T2024)** were tested against sandpaper surfaces of varying roughness (300, 600, and 1200 grit). The maximum spring extension before motion was used to determine friction. Results show that friction increases with surface roughness and varies with material properties, with aluminium exhibiting the highest friction and polycarbonate the lowest. Experimental variability was observed due to surface condition changes and measurement limitations.

1. Introduction

Friction is a fundamental phenomenon in tribology that governs the interaction between contacting surfaces and directly influences the performance, efficiency, and durability of engineering systems. Accurate knowledge of the coefficient of friction under different surface and material conditions is essential for the design and optimization of components such as bearings, sliding interfaces, and mechanical joints. In practical applications, friction is not a constant material property but is highly dependent on factors such as surface roughness, material composition, contact conditions, and environmental effects. Variations in surface condition, particularly roughness, can significantly alter the real area of contact and lead to changes in dominant friction mechanisms such as adhesion, ploughing, and asperity interlocking. One important manifestation of frictional behavior is stick-slip motion. This phenomenon is commonly observed in mechanical systems and is associated with vibration, noise, and potential surface damage. Understanding stick-slip behavior is therefore critical for predicting and controlling friction-induced instabilities. The objective of this study is to experimentally investigate the **coefficient of static friction during stick-slip motion** for different materials under controlled surface roughness conditions. A spring-block system is used to measure the force required to initiate motion, while sandpaper of varying grit sizes is employed to simulate different levels of surface roughness. Through this approach, the study aims to evaluate the combined influence of material properties and surface conditions on frictional behavior.

2. Theory & Background

2.1 Fundamentals of Friction

Friction is the resistive force that opposes relative motion between two contacting surfaces. The magnitude of frictional force is commonly expressed in terms of the coefficient of friction, defined as:

$$\mu = F/N$$

where F is the frictional force and N is the normal reaction force.

In dry contact conditions, friction arises due to a combination of mechanisms including adhesion between surface asperities, ploughing of softer material by harder asperities, and deformation at the contact interface.

2.2 Stick-Slip Motion

Stick-slip motion is a form of frictional instability characterized by alternating phases of sticking and sudden slipping [1]. This behaviour occurs when the maximum static friction exceeds the kinetic friction:

$$\mu_s > \mu_k$$

During the sticking phase, the applied force gradually increases while the body remains stationary. This results in the accumulation of elastic energy within the system, particularly in compliant elements such as springs. When the applied force exceeds the maximum static friction, rapid slipping occurs, leading to a sudden drop in force. This cycle repeats, producing oscillatory motion.

2.3 Spring-Based Force Measurement

In this experiment, the frictional force is determined indirectly using the deformation of a spring. According to Hooke's Law:

$$F = kx$$

where k is the spring constant and x is the extension.

The maximum extension observed just before motion corresponds to the maximum static friction force. Substituting into the friction equation:

$$\mu = kx/N = kx/mg$$

2.4 Effect of Surface Roughness

Surface roughness plays a critical role in frictional behavior by influencing the real area of contact between surfaces. Rough surfaces, such as low-grit sandpaper, exhibit higher friction due to increased asperity interaction, mechanical interlocking, and ploughing effects.

In contrast, smoother surfaces reduce these interactions, leading to lower frictional resistance. Therefore, friction is strongly dependent on both material properties and surface topography.

3. Numerical Analysis

To further understand the behavior of the system, a numerical simulation of the block–surface interaction was performed. A simplified 3D model was used to visualize stress distribution at the contact interface under loading conditions. The simulation indicates that stress concentration is highest during the sticking phase, where the block resists motion and the applied force increases. As the system transitions to slipping, a rapid release of stress occurs, corresponding to the sudden motion observed experimentally. The stress distribution is observed to be non-uniform, with higher concentrations localized near the edges of the contact region. This indicates that the load is not evenly distributed across the interface, and contact occurs primarily through localized asperity interactions. Such edge-dominated stress behaviour is characteristic of real contact conditions in dry friction systems. This behavior aligns with the theoretical concept of stick-slip motion, where energy accumulates during sticking and is released during slipping. The simulation provides qualitative validation of the experimental observations and supports the interpretation of frictional variations across different materials and surface conditions.

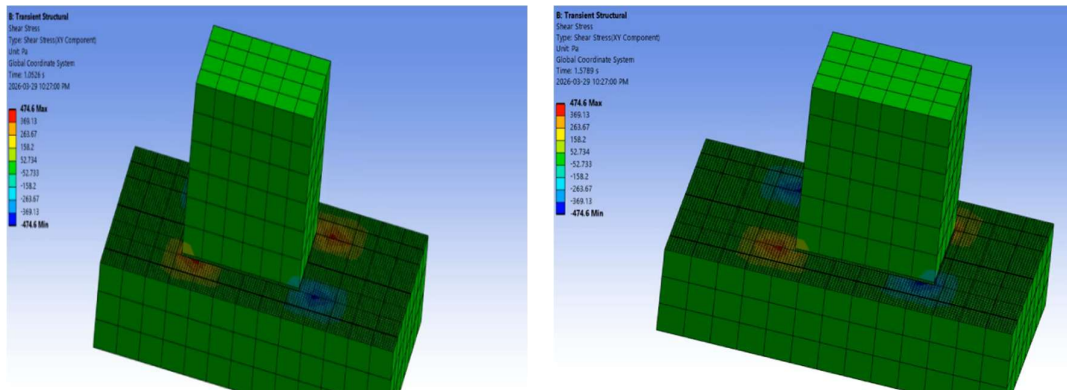


Fig 1: Numerical Simulation of Stress Distribution at Block–Surface Interface during Stick–Slip

4. Experimental Setup

The experimental setup was designed to investigate the coefficient of static friction under stick-slip motion using a spring-block system. The configuration consists of a test specimen placed on a controlled rough surface and connected to an extension spring, which enables indirect measurement of the frictional force through its deformation.

The test block is positioned on a horizontal sandpaper surface that serves as the contact interface. Surface roughness is controlled using sandpaper of different grit sizes, namely **300 grit** (coarse), **600 grit** (medium), and **1200 grit** (fine). A known normal load is applied vertically using calibrated weights to ensure consistent contact conditions between the block and the surface.

The block is connected to an extension spring, which is attached to a pulling mechanism that applies a gradually increasing horizontal force. As the force increases, the spring elongates until the applied force reaches the maximum static friction, at which point the block transitions from sticking to slipping. The maximum extension of the spring at this instant is recorded for analysis.

The extension spring used in this study has a spring constant of **350 N/m**, a free length of **80 mm**, and a maximum extension capacity of **75 mm**. The displacement of the spring is measured using a ruler, allowing the corresponding frictional force to be determined from the observed deformation.

Five different materials were selected as test specimens: **Titanium 6Al-4V**, **Stainless Steel**, **Polycarbonate**, **Inconel 625**, and **Aluminum T2024**. These materials were chosen to represent a range of mechanical and tribological properties, enabling comparison of frictional behavior across different material types.

Initially, the experimental design incorporated a motor-pulley system to provide a controlled and continuous pulling force. However, due to practical challenges in implementation and maintaining system stability, the motorized mechanism could not be successfully integrated into the final setup.

Consequently, a manual hand-crank approach was adopted to apply the pulling force. This method allowed for better control over the rate of force application and facilitated clear observation of the stick-slip transition. Although manual operation introduces minor variability, it provided a simple and effective solution within the available resources. In addition, a control experiment was conducted using a titanium-on-titanium contact configuration to provide a reference for comparison with literature values and to assess the reliability of the experimental setup.

Overall, the setup was designed to ensure simplicity, repeatability, and controlled variation of key parameters such as material type and surface roughness.

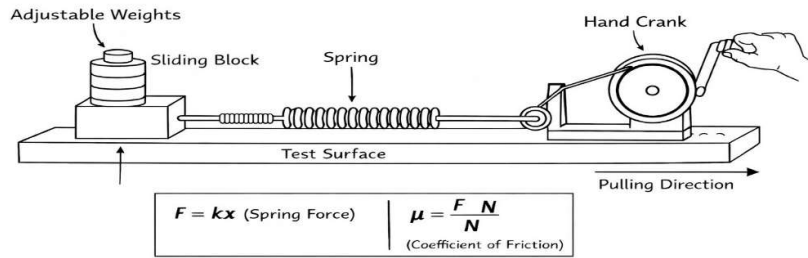


Fig.2: Rough Sketch of Our Experimental Setup (AI Generated)



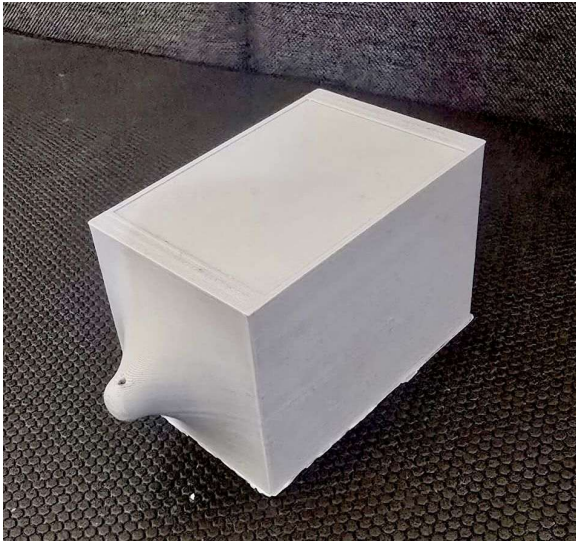
Fig.3: Real Life Experimental Setup



Fig.4: Materials Tested in the Friction Experiment



Fig.5: Calibrated Weights Used to Apply Normal Load



6(a)



6(b)

Fig 6: (a) Sliding Block Used in the Experiment, (b) Detailed View of Material Holding Slot for Specimen Placement

4.1 Manufacturing and acquisition of materials

The experimental rig was constructed using readily available materials to ensure cost-effectiveness and ease of fabrication. The spring was selected based on its stiffness and allowable extension range and was custom designed by McMaster-Carr to meet these specific requirements. Additional components, including the wooden base, sandpapers, and test specimens, were sourced from local hardware stores. Certain elements of the setup, such as the block and gear components, were designed using SolidWorks and subsequently fabricated through 3D printing using university facilities. The overall setup required minimal fabrication effort. No advanced manufacturing techniques were necessary, aligning with the project objective of developing a simple, manually constructed experimental rig.

5. Methodology

For each test, the selected material sample was placed on sandpaper of a specified grit size (300, 600, or 1200). A known normal load was applied through the weight of the sample, ensuring consistent contact conditions.

The sample was connected to an extension spring, and a gradually increasing horizontal force was applied manually using a hand-crank mechanism. The spring extended as the applied force increased. The displacement of the spring was recorded at the instant just before the onset of motion, corresponding to the maximum static friction force.

The applied force was calculated using Hooke's law, and the coefficient of static friction was determined using the ratio of the frictional force to the normal force.

To ensure repeatability and evaluate surface condition effects, three trials were conducted for each test condition:

- **Trial 1: Polished material surface on specified sandpaper**
- **Trial 2: Same polished material repeated under identical conditions.**
- **Trial 3: Unpolished material on new sandpaper of the same grit**

For clarity, the term “trial” used in subsequent tables and figures refers to the predefined test conditions described here, including variations in surface preparation (polished and unpolished) and testing sequence.

This testing approach allowed the influence of surface preparation, repeatability, and sandpaper condition to be systematically evaluated. The procedure was repeated for all materials and grit sizes to obtain a comprehensive dataset for analysis.

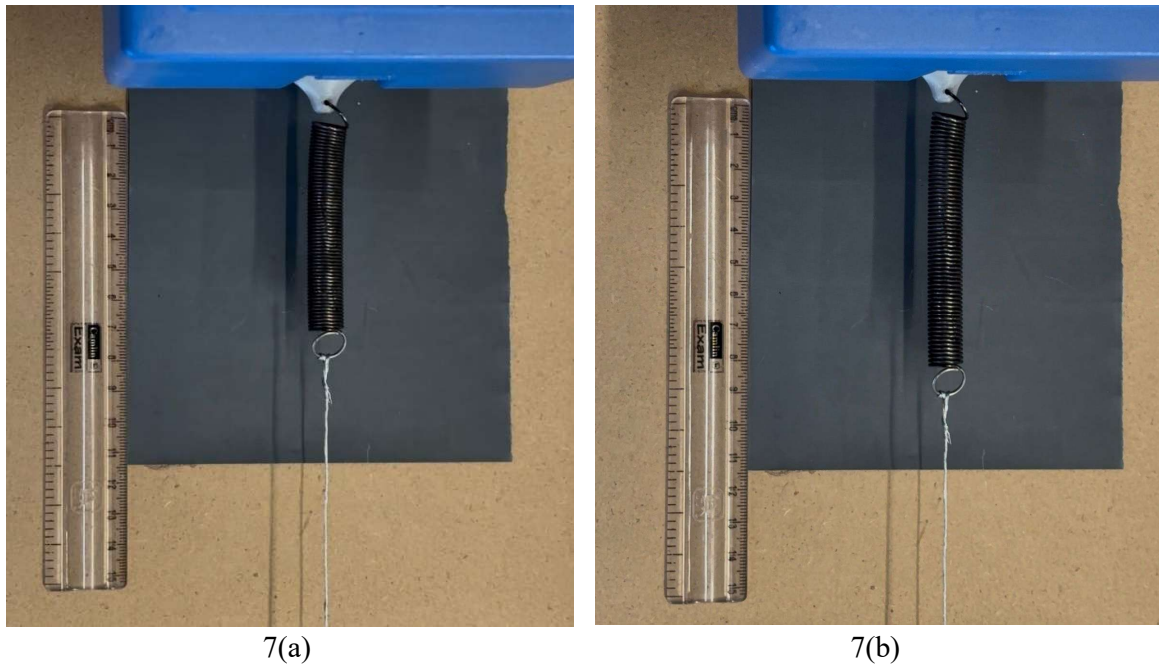


Fig 7: (a) Spring at Rest Position before Loading, (b) Spring Elongation at Maximum Static Friction

5.1 Calibration and Validation

A control test was conducted using a **titanium-on-titanium** contact configuration to assess the accuracy of the experimental setup. The measured coefficient of static friction was approximately **0.43**, compared to a literature value of **0.36** [2][3]. This corresponds to a deviation of about 20%, which is considered acceptable given the simplicity of the setup and manual operation. The difference is attributed to variations in surface condition, repeated testing, and experimental handling. Overall, the results show reasonable agreement with literature, validating the reliability of the setup for comparative analysis.

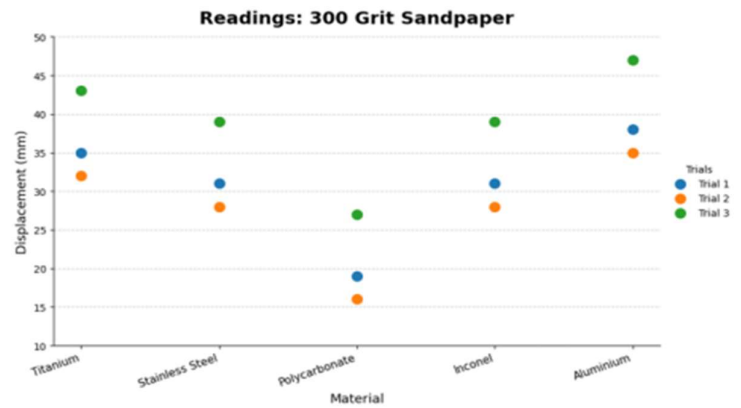
6. Experimental Readings

The experimental readings consist of the measured spring displacement at the onset of motion for each test condition. These values represent the maximum static friction force through their proportional relationship with the spring force. Data was recorded for all combinations of material type and surface roughness (300, 600, and 1200 grit), with three trials conducted per condition. The results are presented in tabulated form, with displacement values reported to two decimal places for consistency.

The data is organized to clearly distinguish between test conditions, including surface preparation (polished and unpolished) and sandpaper condition (reused and new). This allows direct comparison of repeatability effects and surface influence across trials. The recorded displacement values show a consistent pattern, with higher values observed for coarser surfaces and lower values for finer surfaces. These readings form the basis for calculating the coefficient of static friction and are further analyzed in the following section.

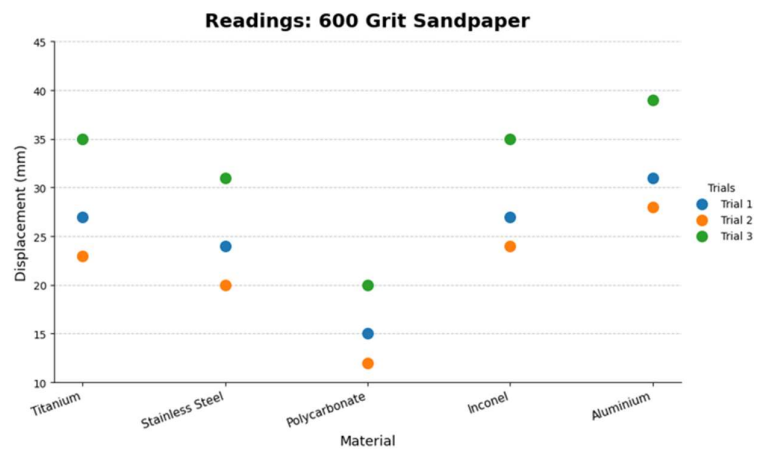
6.1 Measured Displacement for 300 Grit Sandpaper Surface

Material	Trial 1 (mm)	Trial 2 (mm)	Trial 3 (mm)	Average (mm)
Titanium 6AL-4V	35	32	43	36.7
Stainless Steel	31	28	39	32.7
Polycarbonate	19	16	27	20.7
Inconel	31	28	39	32.7
Aluminum	38	35	47	40.0



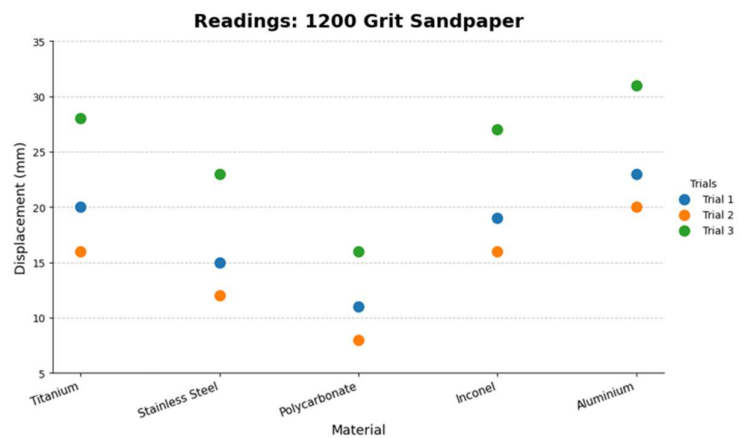
6.2 Measured Displacement for 600 Grit Sandpaper Surface

Material	Trial 1 (mm)	Trial 2 (mm)	Trial 3 (mm)	Average (mm)
Titanium 6AL-4V	27	23	35	28.3
Stainless Steel	24	20	31	25.0
Polycarbonate	15	12	20	15.7
Inconel	27	24	35	28.7
Aluminum	31	28	39	32.7



6.3 Measured Displacement for 1200 Grit Sandpaper Surface

Material	Trial 1 (mm)	Trial 2 (mm)	Trial 3 (mm)	Average (mm)
Titanium 6AL-4V	20	16	28	21.3
Stainless Steel	15	12	23	16.7
Polycarbonate	11	8	16	11.7
Inconel	19	16	27	20.7
Aluminum	23	20	31	24.7



6.4 Calculated Coefficient of Friction

The experimental readings were processed in terms of the coefficient of static friction, calculated from the measured spring displacement at the onset of motion. The applied force was obtained using Hooke's law, and the coefficient of static friction was determined using the relation $\mu = kx/mg$.

Data was collected for all materials across three surface roughness levels with three trials conducted per condition. The dataset is structured to distinguish between trial conditions, including polished and unpolished surfaces, allowing comparison of repeatability and surface effects. The calculated values of the coefficient of static friction for all test conditions are presented in the table below.

Material	300 Grit Trial 1	300 Grit Trial 2	300 Grit Trial 3
Titanium	0.624	0.571	0.767
Stainless Steel	0.553	0.499	0.695
Polycarbonate	0.338	0.285	0.481
Inconel	0.553	0.499	0.695
Aluminium	0.677	0.624	0.838
	600 Grit Trial 1	600 Grit Trial 2	600 Grit Trial 3
Titanium	0.482	0.41	0.624
Stainless Steel	0.428	0.356	0.553
Polycarbonate	0.267	0.214	0.356
Inconel	0.481	0.428	0.624
Aluminium	0.553	0.499	0.695
	1200 Grit Trial 1	1200 Grit Trial 2	1200 Grit Trial 3
Titanium	0.356	0.285	0.499
Stainless Steel	0.267	0.214	0.41
Polycarbonate	0.196	0.142	0.285
Inconel	0.338	0.285	0.482
Aluminium	0.41	0.356	0.553

Fig 8: Coefficient of Static Friction for Different Materials

7. Results and Discussion

The average coefficient of static friction for each material across different surface roughness conditions is shown in Fig 9. Due to the limited availability of literature specifically addressing material-sandpaper interfaces under stick-slip conditions, generalized friction coefficient values from prior studies were considered only as a reference. Based on this comparison, the experimentally obtained values were found to be within a reasonable range, supporting the overall validity of the results.[4][5]

The results indicate a strong dependence of friction on surface roughness. For all materials tested, the coefficient of static friction was highest for the 300 grit surface, followed by 600 grit, and lowest for the 1200 grit surface. This trend confirms that friction increases with increasing surface

roughness. Coarser surfaces contain more pronounced asperities, which enhance mechanical interlocking and ploughing at the contact interface, thereby increasing resistance to motion.

Polished samples consistently showed lower friction values compared to unpolished samples under the same grit conditions. This is due to the smoothing of surface asperities during polishing, which reduces resistance to sliding. In addition, a small reduction in friction was noted in subsequent trials. This behaviour can be explained by changes in the surface condition during testing, such as minor wear or smoothing of the sandpaper and material surface, leading to reduced asperity interaction.

Material-dependent behaviour was also observed across the dataset. Aluminium showed relatively higher friction values compared to the other materials, while polycarbonate exhibited lower friction. These differences can be due to variations in material properties such as hardness, and adhesion characteristics, which influence how the surfaces interact under load.

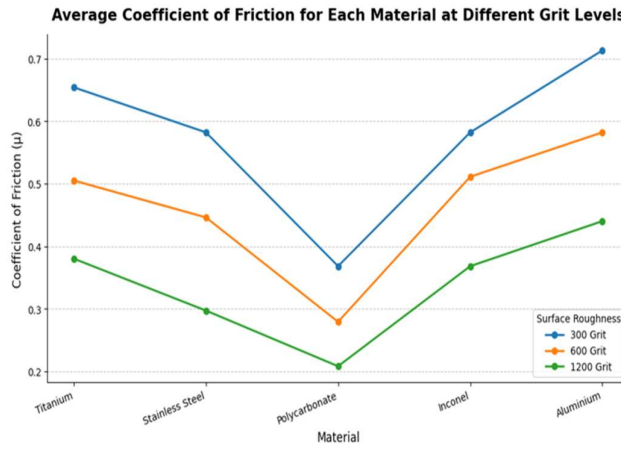


Fig 9: Variation of Average Coefficient of Friction for Different Materials at Various Sandpaper Grit Levels

8. Error Analysis and Limitations

The uncertainty in the calculated coefficient of friction arises from the parameters used in the relation $\mu = kx/mg$, specifically displacement (x), spring constant (k), and mass (m). The combined uncertainty can be expressed as:

$$\frac{\Delta\mu}{\mu} = \sqrt{\left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta k}{k}\right)^2 + \left(\frac{\Delta m}{m}\right)^2}$$

Among these, displacement contributes the most to the overall uncertainty due to manual measurement. The spring constant is assumed constant, although small deviations may exist, and the uncertainty in mass is comparatively minimal.

Based on these factors and experimental observations, an overall uncertainty of approximately 20% in the coefficient of static friction was considered. MATLAB was used to compute the friction coefficient and associated uncertainties, and results were interpreted with respect to this uncertainty range (referred in *Appendix*).

Several practical factors also contributed to experimental error. The surface condition of the samples changed after polishing and repeated testing, affecting contact behaviour. Additionally, variations in experimental conditions such as alignment, contact quality, and environmental effects introduced inconsistencies in the measured values.

The main limitations of the setup arise from the manual application of force, limited measurement precision, and variability in surface conditions. These discrepancies are primarily due to time and financial constraints. Improvements could include the use of higher precision components, proper calibration of the setup, and better control of experimental conditions to enhance repeatability and accuracy.

9. Cost Analysis

The experimental setup was designed to be cost-effective while maintaining sufficient accuracy for friction measurement. The overall cost includes design, materials, fabrication, and testing efforts.

As shown in Fig 10, the total estimated cost of the setup is approximately \$520. The major contributions come from engineering design time and assembling and testing, followed by material and fabrication costs such as test samples, calibrated weights, and 3D printing.

Description	Cost
Professional Engineer Designing	4 hrs $\times$ \$40 = \$160
Supplies & Materials	Test Coupons = \$40 Machinist Labour: 2 hrs $\times$ \$28 = \$56 Calibrated Weights = \$44 Office Supplies = \$50 3D Printing = \$10
Assembling & Testing	4 hrs $\times$ \$40 = \$160
Total Expenditure	\$520
Sale Price (3$\times$ Cost)	\$1560

Fig 10: Estimated Cost Breakdown of the Experimental Setup

Although the setup was developed using accessible resources, the cost analysis highlights that even a simple experimental rig involves both material and labor considerations. For commercial application, the estimated sale price is approximately three times the total cost, reflecting standard engineering and manufacturing practices.

Conclusion

This study successfully developed a simple, cost-effective experimental setup to measure the coefficient of static friction under stick-slip conditions. The results demonstrate that static friction is strongly influenced by both surface roughness and material properties.

A clear trend was observed across all materials, where the coefficient of static friction decreased with increasing sandpaper grit size, indicating lower friction for smoother surfaces. Material-dependent variations were also identified, with aluminium showing higher friction values and polycarbonate exhibiting lower friction.

From a practical perspective, the findings of this study have direct implications in engineering applications where friction control is critical. Surface finishing techniques, such as polishing, can be effectively used to reduce friction in mechanical systems, while rougher surfaces may be desirable in applications requiring higher grip or resistance to motion.

For future work, the experimental setup can be further improved by higher precision measurement tools and controlled environmental conditions in an automated environment. Such enhancements would enable more accurate and repeatable measurements, as well as the study of additional parameters such as lubrication, temperature effects, and dynamic friction behavior.

Team Contribution and Conflict of Interest

This project was carried out collaboratively by all five team members, namely Anandita Tomar, Sri Vyas Narayan, Naria Noor E Bashar, Sanket Bhartiya, and Prabakaran Santhanagopalan. All members contributed equally to the design of the experimental setup, data collection, analysis, and report preparation.

There is no conflict of interest associated with this work.

References

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Appendix

1. MATLAB code for Error Analysis calculation

```
clc; clear; close all;

k = 350;          % N/m
m = 2;           % kg
g = 9.81;        % m/s^2

% Uncertainties (fixed)
dk = 0.10 * k;    % uncertainty in k
dm = 0.10 * m;    % uncertainty in m
dx_mm = 5;        % <-- fixed displacement uncertainty (mm)

n = input('Enter number of trials: ');

x_mm = zeros(1,n);
mu = zeros(1,n);
dmu = zeros(1,n);

for i = 1:n
    fprintf('\n--- Trial %d ---\n', i);

    x_mm(i) = input('Enter displacement (mm): ');

    % Convert to meters
    x_m = x_mm(i) / 1000;

    % FRICTION CALCULATION
    mu(i) = (k * x_m) / (m * g);

    % UNCERTAINTY CALCULATION
    rel_error = sqrt( (dx_mm/x_mm(i))^2 + (dk/k)^2 + (dm/m)^2 );

    dmu(i) = mu(i) * rel_error;
end

% DISPLAY RESULTS

fprintf('\n==== RESULTS =====\n');
for i = 1:n
    fprintf('Trial %d: Friction Coefficient = %.3f ± %.3f\n', i, mu(i), dmu(i));
end

fprintf('\nAverage Friction Coefficient: %.3f\n', mean(mu));
fprintf('Std deviation: %.3f\n', std(mu));

% Plot 1: mu vs displacement
figure;
errorbar(x_mm, mu, dmu, 'o-', 'LineWidth', 2);
xlabel('Displacement (mm)');
ylabel('Friction Coefficient (\mu)');
title('\mu vs Displacement (with uncertainty)');
grid on;

% Plot 2: mu vs trial
figure;
errorbar(1:n, mu, dmu, 's-', 'LineWidth', 2);
xlabel('Trial Number');
ylabel('Friction Coefficient (\mu)');
title('Friction Coefficient vs Trial');
grid on;
```